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Is the Top 1% Getting Away from Taxes?

Introduction

Our primary question was to understand how the economic quality of life varies for Americans across income levels, ages, and time. This is an important question in economics as there is a lot of political debate over progressive and regressive taxation and how it should be applied to balance inequality among Americans and to benefit the American economy as a whole.

Academic literature on this topic has found that the definition of progressive and regressive taxes is much more murky than expected. It also found that the top 10% has grown much more than the rest of people in the US while literature does disagree by how much. Latest papers have found that increasing taxes decreases income inequality.

Through our empirical assessment of the effect of taxation on the distribution of tax burden over the years on different income groups. We can successfully conclude that, while the wealthiest are holding more value they are also at the same time contributing more to the total tax value of America. Furthermore, while the middle class is taxed at a rate not much lower than the elites, their tax burden is decreasing as time goes.

Survey of Literature

One of the first papers that look at taxes and income inequality is *Income Tax Progression, 1929-48* by Musgrave and Thin. They looked at the effects of pre and post tax income on inequality. They looked at 4 categories to measure the progressiveness of taxes which were average rate progressivity, marginal rate progressivity, liability progressivity, and residual income progressivity. They found that taxes could be considered regressive in one category and progressive in another depending on how the income distribution looked like. This paper is considered very important in tax and income inequality work as much of the vocabulary was established from this paper and its analysis is still used today.

Another important paper is *Income Inequality in the United States. 1913-1998* by Piketty and Saez. They used individual IRS tax return data to make a long term series on top income in the United States. They found a U pattern for top income shares with the share being high before World War II and the Great Depression, fell during the Great Depression, and then rose again in the 1970s. This reduction in top income shares was because of the capital shock from the war combined with progressive taxes to prevent the recovery.

Piketty and Saez alongside Zucman then expanded this idea in *Distributional National Accounts: Methods and Estimates for the United States*. They expanded their analysis on their previous paper by looking at all national income including retained corporate earnings as opposed to just tax returns. They then used this data to make pre and post tax income distributions. This data allowed them to look at the redistributive effects of taxes.

However, in *Income Inequality in the United States: Using Tax Data to Measure Long-Term Trends* by Auten and Splinter offer a differing view. They also used IRS tax return data, but argued that previous work overstated the inequality by excluding dependent and non-resident filers and allocating non-tax income differently.

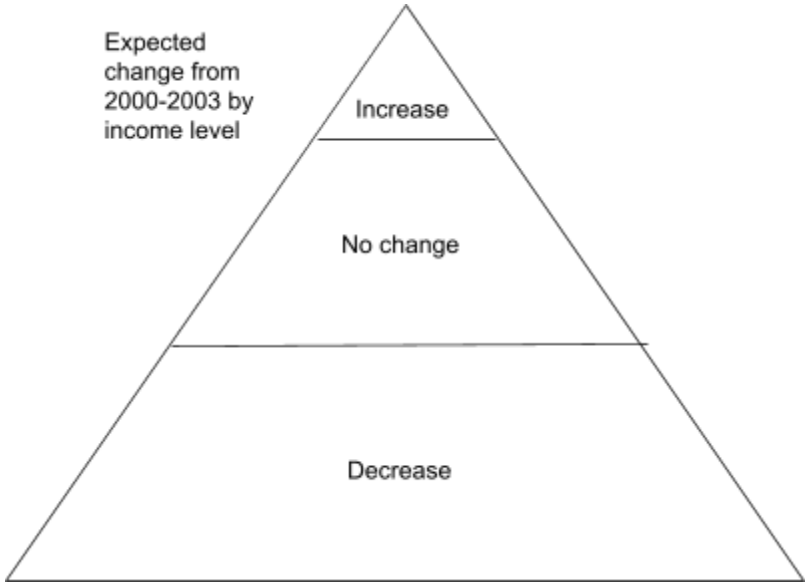
Guillaud, Olckers, and Zemmour also examine this topic in *Income Inequality and Taxes — An Empirical Assessment*. They used a cross country panel data set with fixed effects of country and time and used an instrumental variable of neighboring countries' tax rate to reverse causality between inequality and tax policy. They used the World Income Inequality Database to find that a one percentage point increase in average tax rate was associated with a decline in the Gini coefficient of 0.73 points.

Model Rationale

There exists a pervasive cultural image where the rich enjoy tax breaks, enjoying seaside villas while the poor scramble to make ends meet with what remains of their paycheck after taxes. Since 2000, there have been shifts in national attitudes towards taxation of the wealthy; various legislative reforms have passed with the goal of levying heavier taxes on the upper class. Indeed, the IRS' "sliding scale" imposes a tax rate proportionate to income level (IRS, 2025). The main question was whether tax distribution has become more equitable over time; are the rich contributing more tax than the poor? Analysis from ProPublica indicates the rich are becoming increasingly complacent in their fair share of wealth contribution (Kiel et. al., 2021), so the goal was to see if I could confirm this finding with a different method.

More concretely, the goal was to determine whether the proportion of national tax contributed by each Adjusted Gross Income (AGI) range has changed from 2000-2023.

I hypothesized that if the tax rate has truly become more equitable, then as time goes on, the lower, middle, and upper classes will account for decreasing, roughly equal, and increasing contributions to the national tax generation, respectively (see visualization for rough approximation).



Data

I used Publication 1304 from the IRS to determine tax returns based on individual income where tax generated, in thousands of dollars, was segmented by adjusted gross income (AGI) ranges. Out of 161.3 million Americans, this study uses a stratified probability sample to select 147 individuals' tax returns and estimate the total tax generated for all tax payers that year. I manually compiled the data for the "Tax generated at all rates" variable in Excel for the years 2000-2023, and then uploaded it to Colab for further analysis with Python. The final Excel data looks like this (same pattern all the way to 2023, amounts are in thousands of dollars):

	2000	2001	2002	2003	2004	2005	2006	2007
Under $2,000	53222	26789	16,642	11,228	9,080	10715	10867	12262
$2,000 under $4,000	175650	70292	45,161	30,903	30,383	37480	52708	55576
$4,000 under $6,000	345189	151023	102,949	88,988	78,535	84193	83024	82659
$6,000 under $8,000	660409	355450	294,062	254,546	237,820	229186	238533	237433
$8,000 under $10,000	1114373	777835	545,140	534,454	495,293	442307	406204	383317
$10,000 under $12,000	1701847	1365943	982,344	932,199	845,086	797709	754870	742920
$12,000 under $14,000	2419538	2131858	1,421,976	1,325,606	1,225,668	1178715	1193483	1119567
$14,000 under $16,000	3341833	3130308	2,070,601	1,898,310	1,734,325	1671832	1609722	1574068
$16,000 under $18,000	4249634	3950516	2,817,446	2,441,156	2,406,610	2267712	2140998	1967035
$18,000 under $20,000	5293477	4769242	3,596,051	3,169,157	3,041,159	2900117	2795763	2686445
$20,000 under $25,000	16109360	15269866	11,582,249	10,631,719	10,242,795	9888565	9784387	9410901
$25,000 under $30,000	19046013	18946371	15,079,322	13,880,145	13,603,201	13592847	13325950	13213065
$30,000 under $40,000	44307242	43882869	37,324,746	34,725,739	33,865,146	32844794	32970635	33595654
$40,000 under $50,000	49596589	49309375	42,612,342	39,416,550	39,622,738	38443590	38102778	38989594
$50,000 under $75,000	123389614	123298530	110,360,137	103,554,424	105,987,893	105556349	107343865	109405381
$75,000 under $100,000	103922256	103648825	98,700,327	90,071,461	94,028,152	94697328	99607990	103614156
$100,000 under $200,000	185016291	187160993	176,419,717	165,058,719	178,200,665	192474765	213889256	234282751
$200,000 under $500,000	144638780	134043009	123,912,146	117,695,811	134,052,737	152646998	168370440	187475287
$500,000 under $1,000,000	75682396	68194292	63,700,073	60,145,607	70,921,028	83871841	92728417	101567991
$1,000,000 under $1,500,000	35307670	30676646	27,802,676	25,758,910	31,285,276	38338032	44152720	48571449
$1,500,000 under $2,000,000	22634618	18741216	16,131,938	15,462,081	19,605,837	24189349	26816518	29625196
$2,000,000 under $5,000,000	58561043	46491534	39,423,204	37,415,357	48,518,222	61785443	72283115	78564014
$5,000,000 under $10,000,000	34570570	24484509	20,389,135	19,637,138	26,124,134	35001479	39972419	44508547
$10,000,000 or more	76502781	45769129	34,500,461	36,169,017	55,073,487	79761612	92648701	113741087

Dollar amounts were adjusted to 2023 levels of inflation.

According to the Pew Research Center, an income in the range of $59,430 to $178,290 is considered "middle class" (2022 estimates adjusted for 2023 inflation)(Fry, 2024). I color coded my IRS AGI classifications roughly following this guideline.

Empirical Application

I. Model, Data Analysis

The goal was to use an interaction model to determine the changes in tax contribution by income level.

$$\text{Prop of National Tax}_{it} = \beta_0 + \beta_1 \text{ Year}_t + \beta_2 \text{ AGI}_i + \beta_3 \text{ Year}_t * \text{AGI}_i + u_{it}$$

B3, the interaction coefficient, tells us what the changes across time are for each AGI and whether such changes are significant.

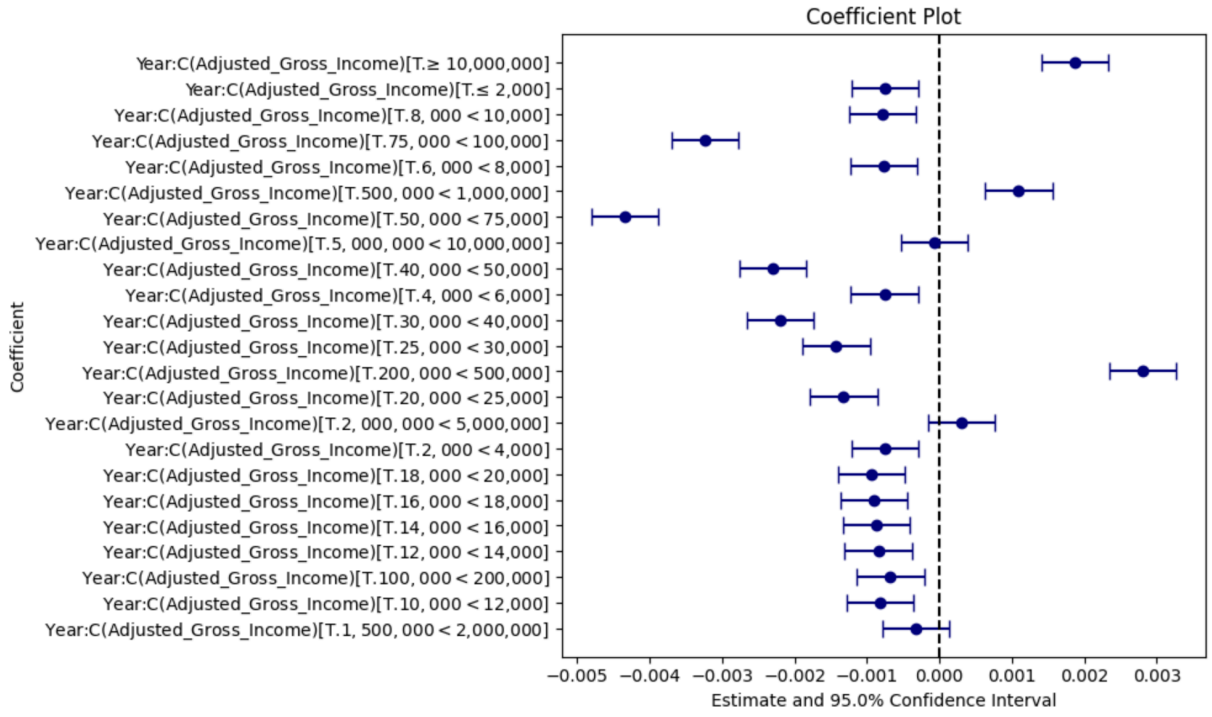
* Hypotheses are aggregated across classes here for simplicity, but each AGI will be evaluated separately.

AGI Range	Hypothesis	p-value	Reject, fail to reject	Conclusion
Under $2,000 $2,000 under $4,000 $4,000 under $6,000 $6,000 under $8,000 $8,000 under $10,000 $10,000 under $12,000 $12,000 under $14,000 $14,000 under $16,000 $16,000 under $18,000 $18,000 under $20,000 $20,000 under $25,000 $25,000 under $30,000 $30,000 under $40,000 $40,000 under $50,000	Ho: B3 > 0 Ha: B3 ≤ 0 "The portion of tax contributed by the lower class has decreased over time"	All < 0.05	Reject Ho for all.	As time goes on, tax contribution decreases for the lower class.
$50,000 under $75,000 $75,000 under $100,000 $100,000 under $200,000	Ho: B3 = 0 Ha: B3 ≠ 0 "The portion of tax contributed by the middle class has not changed over time"	All < 0.05	Reject Ho for all.	As time goes on, tax contribution changes (actually decreases) for the middle class.
$200,000 under $500,000 $500,000 under $1,000,000 $1,000,000 under $1,500,000 $1,500,000 under $2,000,000 $2,000,000 under $5,000,000 $5,000,000 under $10,000,000 $10,000,000 or more	Ho: B3 < 0 Ha: B3 ≥ 0 "The portion of tax contributed by the upper class has increased over time"	> 0.05 for $1.5 - $10 million AGI brackets, < 0.05 for all remaining, but some are in a positive direction	Reject Ho only for [$200k, $500k), [$500k, $1m), [$10m, $10m+) brackets. Fail to reject all remaining.	As time goes on, tax contribution increases significantly only for [$200k, $500k), [$500k, $1m), [$10m, $10m+) brackets. Significant decrease for [$1m, $1.5m).

II. Discussion of Results

For the lower and middle classes, the contribution to national tax decreases. For upper classes, this contribution increases only for the lower and upper ends of this range. There is no significant increase for the middle areas of the upper class, and there is actually a decrease in contribution for the $1m - $1.5m income range. At face value, the immediate takeaway is that lower and middle classes face lessening tax burdens as certain segments of the upper classes compensate by increasingly accounting for the national tax collection. This could be considered equitable looking across classes, but within the millionaire income bracket, it appears some of the rich are in fact lagging in their contributions. It could be said that only some rich people are appropriately compensating for their poorer fellow Americans.

Lower and middle class all have slightly lower coefficients, while the upper class AGI ranges have much bigger jumps. Most likely this is due to levying a bigger portion of the tax burden to a smaller group of earners. Interestingly enough, the $200k - $500k bracket has one of the biggest leaps of all. Further analysis can help explain the magnitude of such changes; it could be factors like the expansion of the $200k - $500k income class or the shrinking of the lower-millionaire class.



III. Limitations

There are some significant limitations to the interpretation of these findings. The IRS data are based on a small sample relative to the total population; although the sample selection method allows for more segment-accurate estimates, it can only capture a limited amount of information and variability.

My model is also likely to be flawed. Though my adjustment for inflation adds an element of a time fixed effect, I was unable to add time and AGI (entity) fixed effects; I tried multiple fixed effects OLS models but feols was the only one compatible with the structure of my data frame, and the limited number of independent variables severely limited my ability to add fixed effects. Thus, my B3 estimates are likely heavily biased as there are time varying elements, such as economic crises (e.g. 2008, COVID), and AGI varying factors (e.g. a new tax on millionaires, offshore bank accounts as a tax evasion strategy) that will certainly affect the calculated coefficient. These omitted variables could also explain the decreasing contribution of the $1m - $1.5m income bracket.

Limitations aside, the findings from this analysis still have utility in the sense that they give a general, macro-level overview of how contributions to national tax collection have evolved for different income levels over time. Additionally, the IRS is likely to be an authoritative source for tax data, as all Americans are required to file through them, allowing them to amass a comprehensive dataset. It can be assumed that this credibility and representation are reflected in the data used in this regression.

Political reforms are most likely to be a significant confounder for this analysis. My model can only describe association, but including a variable for legal changes will likely provide a more accurate illustration of the precise changes affecting each AGI's tax contribution.

Conclusion

As time goes on, richer Americans are increasingly responsible for the national tax collected that year. This can support the original claim that the US tax system is becoming more equitable over time. However, within this upper class, some are carrying the burden to a greater extent, implying the possibility for further analysis.

Sources

"Are You in the American Middle Class?" Pew Research Center, 16 Sept. 2024, www.pewresearch.org/short-reads/2024/09/16/are-you-in-the-american-middle-class/. Accessed 12 May 2026.

Auten, Gerald, and David Splinter. "Income Inequality in the United States: Using Tax Data to Measure Long-Term Trends." Journal of Political Economy, vol. 132, no. 7, 2024, pp. 2179–2227.

Guillaud, Elvire, Matthieu Olckers, and Michaël Zemmour. "Income Inequality and Taxes — An Empirical Assessment." World Inequality Lab Working Paper, no. 2023/14, 2023.

Musgrave, Richard A., and Tun Thin. Income Tax Progression, 1929–48. National Bureau of Economic Research, 1948.

Piketty, Thomas, and Emmanuel Saez. "Income Inequality in the United States, 1913–1998." Quarterly Journal of Economics, vol. 118, no. 1, 2003, pp. 1–39.

"Real Median Personal Income in the United States." Federal Reserve Bank of St. Louis, FRED, FRED Economic Data Series MEPAINUSA672N. Accessed 11 May 2026.

United States, Internal Revenue Service. Individual Complete Report (Publication 1304), Table 3.5, "Returns with Modified Taxable Income: Tax Generated." IRS Publication 1304 PDF. Accessed 12 May 2026.

Woolhouse, Megan Rose. "You May Be Paying a Higher Tax Rate Than a Billionaire." ProPublica, ProPublica, 7 Oct. 2021. Accessed 12 May 2026.